AIPro iNet Solution Guide for Ubuntu Nvidia GPU v1.1 (English)

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General information

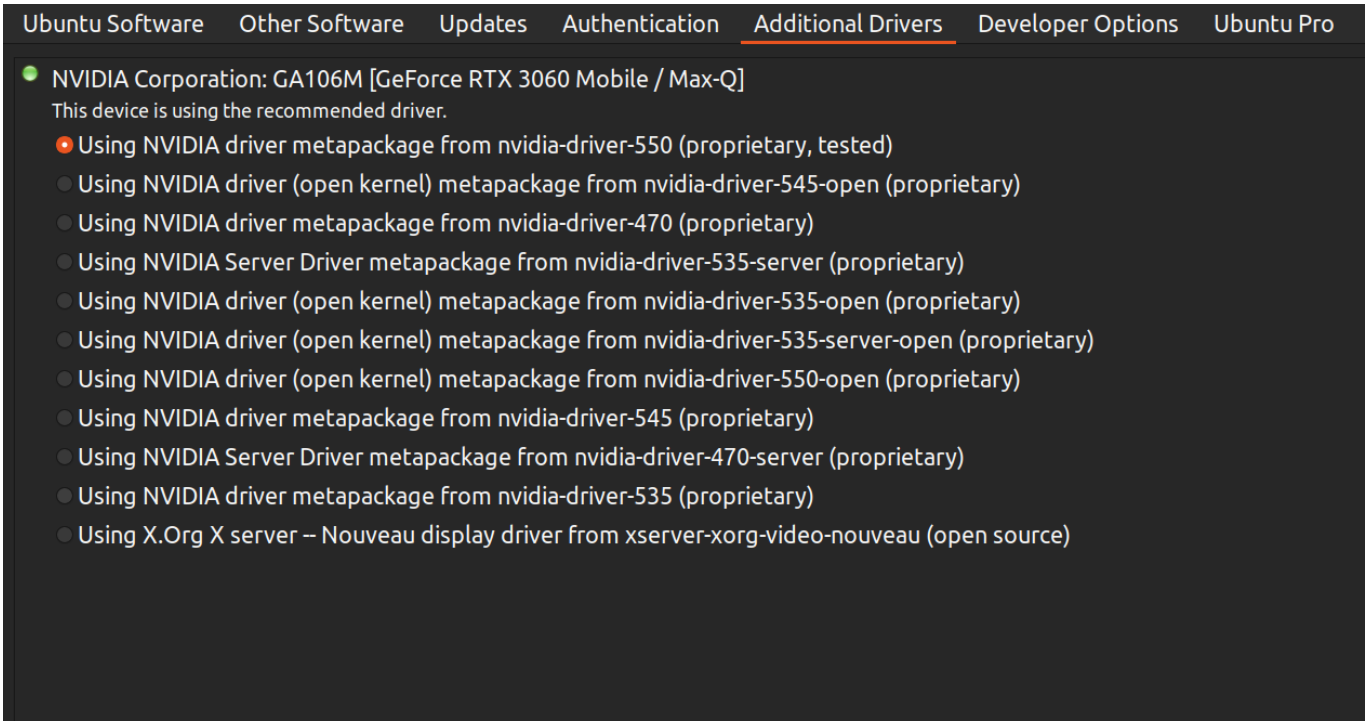
Source code and documents

- Link: AIProCo/iNet-API-Demo
- This guild file could found in iNet source code at `docs`

Dev environments (envs)

- Ubuntu 22.04 LTS
  - You can install Ubuntu using this guide
  - You can install Nvidia driver using this guide
- Cuda 12.1.0 (need to install)
- cuDNN 8.9.6 (need to install)
- OpenCV-4.9.0 (included)
- TensorRT-8.6.1.6 (included)
- OpenVINO 2023.2 (included)
- GCC 13 (C++ complier) required due to iNet used `std::format` in C++ 20 standards
- C++ build tool `CMake` with minimum VERSION 3.10 (tested with version 3.22.1 in this guide)

The following image is the test environment for this guide:



Technical Guides

> **Notice**:
>
> - All paths in this guide is relative path to `iNet-API-Demo` folder.
> - BEFORE running a script file (.sh), make sure it allowed to run by execute this command: `sudo chmod +x script.sh`
> - when run CMake, make sure you specify gcc-13: `cmake -S . -B build -DCMAKE_C_COMPILER=/usr/bin/gcc-13 -DCMAKE_CXX_COMPILER=/usr/bin/g++-13 ...<other_args>...`

Step 1 : Install C++ complier GCC 13 and dependencies

- original guide
- run these cmd:

```
sudo add-apt-repository ppa:ubuntu-toolchain-r/test
sudo apt-get update
sudo apt-get install gcc-13 g++-13
# test
gcc-13 --version
```

- Install dependent libs: note that tbb (Intel oneAPI Threading Building Blocks) may not be installed by default, so you need to install it manually.

```
sudo apt install cmake libtbb2 g++ wget unzip ffmpeg libgtk2.0-dev
libavformat-dev libavcodec-dev libavutil-dev libswscale-dev libtbb-dev
libjpeg-dev libpng-dev libtiff-dev
```

Step 2: Install Cuda 12.1 for Ubuntu 22.04

- Install Cuda by following this official video CUDA Tutorials I Installing CUDA Toolkit on Windows and WSL - YouTube with the `.run` file downloaded from CUDA Toolkit Archive.
    - Note that if we already install Nvidia driver (using `Additional Drivers` in Ubuntu), we should **NOT** install Nvidia driver again when installing Cuda Toolkit.
- Add Cuda to system `PATH` and `LD_LIBRARY_PATH`

```
export PATH=/usr/local/cuda-12.1/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda-12.1/lib64:$LD_LIBRARY_PATH
```

Step 3: Install cuDNN 8.9.6 using Tar file

- Install `cuDNN` using `.tar` file with this guide where `.tar` file can be downloaded at cuDNN Archive

Choose `Download cuDNN v8.9.6 (November 1st, 2023), for CUDA 12.x` and download `cuDNN Library for Linux x86_64 (Tar)`.

```
tar -xvf cudnn-linux-x86_64-8.9.6.50_cuda12-archive.tar.xz
sudo cp cudnn-*-archive/include/cudnn*.h /usr/local/cuda-12.1/include
sudo cp -P cudnn-*-archive/lib/libcudnn* /usr/local/cuda-12.1/lib64
sudo chmod a+r /usr/local/cuda/include/cudnn*.h /usr/local/cuda-
12.1/lib64/libcudnn*
```

Step 4: Download and extract necessary files

> **NOTE**: the `.net` models in `inputs` folder are used for linux environment, so **DO NOT USE* the `.net` models in `inputs` folder for Windows environment.

- Download and upzip one of the followings zip files. Each zip files has 3 directories after unzip: `bin`, `inputs`, and `videos`. Then, copy and paste `bin`, `inputs`, and `videos` directories to the solution directory (the directory including the .sln file).
  - Cuda compute capability of your GPU should be 8.6 (RTX-30xx) or 8.9 (RTX-40xx):
    - RTX-30xx
    - RTX-40xx

Step 8: Build and run iNet framework

> **NOTE**: If you want to build for `CPU`, the flag in `CMakeLists.txt` must be turn ON `set (BUILD_FOR_CPU ON)`

- Check and modify the contents of the `CMakeLists.txt` files if needed. Make sure configs correct, especially for *header file directories* and *lib file directories*.

Two methods to build the project, and run the executable file:

- Run `build.sh [-force] [-debug]` to build or rebuild from scratch [-force] and enable debug mode [-debug].
  - Note that the executable file (`client`) are automatically copied to `bin` folder after build process.
    - Move `inputs` folder (with `config.json` and `.net` mode files) and `videos` folder are moved/copied into `bin` folder.
    - Run `client`:
      - `cd bin`
      - `./client`
- Method 2: Simply run the script `run.sh -b` to build and run the executable file.